



Python Programming Intern

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Project no. : Forth

Project name: General Knowledge Quiz

General Knowledge Quiz

Application

Introduction:

General Knowledge Quiz Game is a simple command-line application developed using Python. The program presents the user with a series of multiple-choice general knowledge questions. The user selects an answer from the given options, and the program checks whether the selected answer is correct.

At the end of the quiz, the program displays the user's answers, guesses, total score, and percentage.

This project was developed as part of the Python Programming Internship at DecodeLabs.

Tools & Technologies Used:

- VS Code
- Python
- GitHub

Objectives of the Project:

The main objectives of the General Knowledge Quiz Game are:

- To develop a simple quiz game using Python.
- To practice Python programming fundamentals.
- To learn how to work with tuples and nested tuples.
- To understand the use of loops and conditional statements.
- To practice taking and processing user input.
- To calculate and display the user's score and percentage.
- To gain hands-on experience in developing a command-line application.

Features:

Following are the features of the General Knowledge Quiz Game:

1. Contains 10 general knowledge questions.
2. Provides four multiple-choice options for each question.
3. Allows the user to select an answer using A, B, C, or D.
4. Automatically checks whether the selected answer is correct.
5. Displays "CORRECT!" for a correct answer.
6. Displays "WRONG!" and the correct answer for an incorrect response.
7. Keeps track of the user's guesses.
8. Calculates the total score and percentage.
9. Displays all correct answers and the user's guesses at the end.
10. Simple and easy-to-use command-line interface.

Python Concepts Used:

Following fundamental Python Concepts are used in this project:

- Variables

```
answers = ("C", "B", "A", "  
guesses = []  
score = 0  
question_number = 0
```

- Tuples and Nested tuples

```
questions = ("How many continents are there in the world? ",  
            "Who is known as the father of computers? ",  
            "What is the Chemical Symbol for Water? ",  
            "Which is the largest planet in our solar system? ",  
            "Which language is primarily used to create the structure of web pages? ",  
            "Which is the fastest land animal? ",  
            "What is the capital of France? ",  
            "How many players are there in a football team? ",  
            "How many bones are there in the human body? ",  
            "Which planet is the hottest in our solar system? ")  
  
options = ( ("A) 5", "B) 6", "C) 7", "D) 8" ),  
            ("A) Alan Turing", "B) Charles Babbage", "C) John von Neumann", "D) Bill Gates" ),  
            ("A) H2O", "B) CO2", "C) NaCl", "D) O2" ),  
            ("A) Jupiter", "B) Saturn", "C) Uranus", "D) Neptune" ),  
            ("A) JavaScript", "B) Python", "C) HTML", "D) CSS" ),  
            ("A) Cheetah", "B) Lion", "C) Horse", "D) Dog" ),  
            ("A) London", "B) Berlin", "C) Paris", "D) Rome" ),  
            ("A) 10", "B) 11", "C) 12", "D) 13" ),  
            ("A) 206", "B) 205", "C) 201", "D) 208" ),  
            ("A) Venus", "B) Mercury", "C) Mars", "D) Jupiter" )  
  
answers = ("C", "B", "A", "A", "C", "A", "C", "A", "B")
```

- User input using "input()"

```
guess = input("Enter (A, B, C, D): ").upper()  
guesses.append(guess)
```

- String methods using "upper()"

```
guess = input("Enter (A, B, C, D): ").upper()  
guesses.append(guess)
```

- "for" loops

```
print("Answers: ", end="")  
for answer in answers:  
    print(answer, end=" ")  
print()  
  
print("Your guesses: ", end="")  
for guess in guesses:  
    print(guess, end=" ")  
print()
```

- "if-else" conditional statements

```
if guess == answers[question_number]:  
    score += 1  
    print("CORRECT!")  
else:  
    print("WRONG!")  
    print(f"{answers[question_number]} is the correct answer.")  
  
question_number += 1
```

- Lists

```
answers = ("C", "  
guesses = []  
score = 0
```

- List method "append()"

```
guess = input("Enter (A, B, C, D) : ")
guesses.append(guess)
```

- "len()" function

```
percentage = score/ len(questions) * 100
print(f"Your percentage is: {percentage}%")
print()
```

- Arithmetic operations

```
percentage = score/ len(questions) * 100
print(f"Your percentage is: {percentage}%")
print()
```

- String formatting using f-strings

```
print(f"Total Correct answers: {score}")

percentage = score/ len(questions) * 100
print(f"Your percentage is: {percentage}%")
print()
```

- "end" parameter in "print()"

```
print("Answers: ", end="")
for answer in answers:
    print(answer, end=" ")
print()

print("Your guesses: ", end="")
for guess in guesses:
    print(guess, end=" ")
print()
```

Working of the Program:

The execution process of the program is as follows:

1. A tuple containing 10 general knowledge questions is created and a nested tuple containing four options for each question is created.

```
questions = ("How many continents are there in the world? ",
             "Who is known as the father of computers? ",
             "What is the Chemical symbol for Water? ",
             "Which is the largest planet in our solar system? ",
             "Which language is primarily used to create the structure of web pages? ",
             "Which is the fastest land animal? ",
             "What is the capital of France? ",
             "How many players are there in a football team? ",
             "How many bones are there in the human body? ",
             "Which planet is the hottest in our solar system? ")

options = ( ("A) 5", "B) 6", "C) 7", "D) 8" ),
           ("A) Alan Turing", "B) Charles Babbage", "C) John von Neumann", "D) Bill Gates" ),
           ("A) H2O", "B) CO2", "C) NaCl", "D) O2" ),
           ("A) Jupiter", "B) Saturn", "C) Uranus", "D) Neptune" ),
           ("A) Javascript", "B) Python", "C) HTML", "D) CSS" ),
           ("A) Cheetah", "B) Lion", "C) Horse", "D) Dog" ),
           ("A) London", "B) Berlin", "C) Paris", "D) Rome" ),
           ("A) 10", "B) 11", "C) 12", "D) 13" ),
           ("A) 206", "B) 205", "C) 201", "D) 208" ),
           ("A) Venus", "B) Mercury", "C) Mars", "D) Jupiter" )

answers = ("C", "B", "A", "A", "C", "A", "C", "C", "A", "B")
```

2. Another tuple stores the correct answers for all questions.

```
answers = ("C", "B", "A", "A", "C", "A", "C", "C", "A", "B")
```

3. Empty lists and variables are initialized to store the user's guesses, score, and question number.

```
answers = ( 'C', 'B', 'A',  
guesses = []  
score = 0  
question_number = 0
```

4. The program displays the title of the General Knowledge Quiz Game.

```
print("-----")  
print("General Knowledge Quiz Game")  
print("-----")
```

5. A "for" loop goes through each question one by one. The corresponding options are displayed using the question number. The user is asked to enter an answer from A, B, C, or D. The user's input is converted to uppercase using "upper()". The user's guess is stored in the "guesses" list using "append()". The program compares the user's guess with the correct answer. If the answer is correct, the score is increased by 1 and "CORRECT!" is displayed. If the answer is incorrect, "WRONG!" and the correct answer are displayed. The question number is increased and the program continues to the next question.

```
for question in questions:  
    print("-----")  
    print(question)  
    for option in options[question_number]:  
        print(option)  
    guess = input("Enter (A, B, C, D): ").upper()  
    guesses.append(guess)  
  
    if guess == answers[question_number]:  
        score += 1  
        print("CORRECT!")  
    else:  
        print("WRONG!")  
        print(f"{answers[question_number]} is the correct answer.")  
  
    question_number += 1
```

6. After all questions are completed, the program displays the correct answers and the user's guesses.

```
print()  
print("Quiz completed!")  
print()  
print("-----YOUR RESULT-----")  
print()  
  
print("Answers: ", end="")  
for answer in answers:  
    print(answer, end=" ")  
print()  
  
print("Your guesses: ", end="")  
for guess in guesses:  
    print(guess, end=" ")  
print()  
  
print(f"Total Correct answers: {score}")  
  
percentage = score/ len(questions) * 100  
print(f"Your percentage is: {percentage}%")  
print()
```

7. The score is calculated based on the total number of questions.

```
Answers: C B A A C A C C A B  
Your guesses: C B A A C A C C D B  
Total Correct answers: 9  
Your percentage is: 90.0%
```

8. The percentage is calculated and displayed to the user.

```
Answers: C B A A C A C C A B
Your guesses: C B A A C A C C D B
Total Correct answers: 9
Your percentage is: 90.0%
```

Application Flow:

The overall basic workflow of the application is as follow:

Start → Display "General Knowledge Quiz Game" → Display a question → Display four answer options → Ask the user to enter A, B, C, or D → Store the user's guess → Compare the guess with the correct answer → Display CORRECT or WRONG → Move to the next question → Repeat until all questions are completed → Calculate score → Calculate percentage → Display results → END.

Code:

#=====THE GENERAL KNOWLEDGE QUIZ GAME=====

```
questions = ("How many continents are there in the world?", "
    "Who is known as the father of computers? ",
    "What is the Chemical Symbol for Water? ",
    "Which is the largest planet in our solar system? ",
    "Which language is primarily used to create the structure of web pages? ",
    "Which is the fastest land animal? ",
    "What is the capital of France? ",
    "How many players are there in a football team? ",
    "How many bones are there in the human body? ",
    "Which planet is the hottest in our solar system? "
)
```

```
options = ( ("A) 5", "B) 6", "C) 7", "D) 8"),
    ("A) Alan Turing", "B) Charles Babbage", "C) John von Neumann", "D) Bill
Gates"),
    ("A) H2O", "B) CO2", "C) NaCl", "D) O2"),
    ("A) Jupiter", "B) Saturn", "C) Uranus", "D) Neptune"),
    ("A) JavaScript", "B) Python", "C) HTML", "D) CSS"),
    ("A) Cheetah", "B) Lion", "C) Horse", "D) Dog"),
    ("A) London", "B) Berlin", "C) Paris", "D) Rome"),
    ("A) 10", "B) 11", "C) 12", "D) 13"),
    ("A) 206", "B) 205", "C) 201", "D) 208"),
    ("A) Venus", "B) Mercury", "C) Mars", "D) Jupiter" )
)
```

```
answers = ("C", "B", "A", "A", "C", "A", "C", "C", "A", "B")
guesses = []
score = 0
question_number = 0
```

```

print("-----")
print("General Knowledge Quiz Game")
print("-----")

for question in questions:
    print("-----")
    print(question)
    for option in options[question_number]:
        print(option)
    guess = input("Enter (A, B, C, D): ").upper()
    guesses.append(guess)

    if guess == answers[question_number]:
        score += 1
        print("CORRECT!")
    else:
        print("WRONG!")
        print(f"{answers[question_number]} is the correct answer.")

    question_number += 1

print()
print("Quiz completed!")
print()
print("-----YOUR RESULT-----")
print()

print("Answers: ", end="")
for answer in answers:
    print(answer, end=" ")
print()

print("Your guesses: ", end="")
for guess in guesses:
    print(guess, end=" ")
print()

print(f"Total Correct answers: {score}")

percentage = score/ len(questions) * 100
print(f"Your percentage is: {percentage}%")
print()

```

Sample Output:

```
PS C:\Users\amima\OneDrive\Desktop\VS CODE folder> python Quiz_General_Knowledge.py
-----
General Knowledge Quiz Game
-----
How many continents are there in the world?,
A) 5
B) 6
C) 7
D) 8
Enter (A, B, C, D): c
CORRECT!
-----
Who is known as the father of computers?
A) Alan Turing
B) Charles Babbage
C) John von Neumann
D) Bill Gates
Enter (A, B, C, D): b
CORRECT!
-----
What is the Chemical Symbol for Water?
A) H2O
B) CO2
C) NaCl
D) O2
Enter (A, B, C, D): a
CORRECT!
-----
Which is the largest planet in our solar system?
A) Jupiter
B) Saturn
C) Uranus
D) Neptune
Enter (A, B, C, D): a
CORRECT!
-----
Which language is primarily used to create the structure of web pages?
A) JavaScript
```

```
-----
Which language is primarily used to create the structure of web pages?
A) JavaScript
B) Python
C) HTML
D) CSS
Enter (A, B, C, D): c
CORRECT!
-----
Which is the fastest land animal?
A) Cheetah
B) Lion
C) Horse
D) Dog
Enter (A, B, C, D): a
CORRECT!
-----
What is the capital of France?
A) London
B) Berlin
C) Paris
D) Rome
Enter (A, B, C, D): c
CORRECT!
-----
How many players are there in a football team?
A) 10
B) 11
C) 12
D) 13
Enter (A, B, C, D): c
CORRECT!
-----
How many bones are there in the human body?
A) 206
B) 205
C) 201
D) 208
Enter (A, B, C, D): d
```

```
Enter (A, B, C, D): c
CORRECT!
-----
How many bones are there in the human body?
A) 206
B) 205
C) 201
D) 208
Enter (A, B, C, D): d
WRONG!
A is the correct answer.
-----
Which planet is the hottest in our solar system?
A) Venus
B) Mercury
C) Mars
D) Jupiter
Enter (A, B, C, D): b
CORRECT!

Quiz completed!

-----YOUR RESULT-----

Answers: C B A A C A C C A B
Your guesses: C B A A C A C C D B
Total Correct answers: 9
Your percentage is: 90.0%

PS C:\Users\amima\OneDrive\Desktop\VS CODE folder> 
```

Testing:

The application was tested by performing each operation:

Sr.	Task	Result	Status
1.	Correct answers	Correct answers are identified and score increases	Passed
2.	Incorrect answers	Wrong answers are identified and correct answers are displayed	Passed
3.	Different answer choices (A, B, C, D)	Program accepts and checks the selected option.	Passed

Conclusion:

The General Knowledge Quiz Game successfully demonstrates the practical use of basic Python programming concepts. The project helped in understanding tuples, nested tuples, lists, loops, conditional statements, user input, indexing, and arithmetic operations.

GitHub Repository:

[AmimaBatoool/Task-4 AmimaBatoool: A Python-based General Knowledge Quiz Game with multiple-choice questions, answer checking, score calculation, and percentage results.](#)

Project status: Completed.